The NTSB Investigative Process

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NTSB Primer

- Established in 1967 as part of the Department of Transportation
  - 1975 became fully independent
  - Five board members are Presidential appointments (5 yr terms)
  - ~400 employees
- Investigates ~2500 accidents per year (link)
  - Aviation: ~2000; Marine, rail, highway, pipeline: ~500
- "Accident/incident investigations are fact-finding proceedings with no formal issues and no adverse parties … and are not conducted for the purpose of determining the rights or liabilities of any person."
  - Title 49 *Code of Federal Regulations* section 831.4

NTSB Mission

- Maintaining our congressionally mandated independence
- Conducting objective, thorough investigations and safety studies
- Deciding fairly and objectively appeals of enforcement actions by the FAA and US Coast Guard and certificate denials by the FAA
- Advocating for implementation of safety recommendations
- Assisting victims and survivors of transportation disasters and their families

From "About the NTSB"

Selecting Accidents to Investigate

- NTSB's responsibility to investigate accidents varies by transportation mode
  - NTSB is required to investigate all civil aviation accidents, and all railroad and pipeline accidents that result in fatalities or substantial property damage.
  - For highway crashes, NTSB may initiate an investigation at its discretion.

- In determining whether to investigate, NTSB considers:
  1. Are there national safety issues as opposed to site-specific issues?
  2. Does the crash involve emerging technologies or public safety issues?
  3. Is there potential for safety recommendations?
  4. Does NTSB have the resources (i.e., staff availability and capacity)?

Investigative Process Overview

- Initial notification and decision to investigate
- Gathering facts on-site
- Analysis facts and determination of probable cause
- Acceptance of a final report
- Advocating for acceptance of safety recommendations arising from the investigation
- Invoke subpoena and prosecutorial power

https://www.ntsb.gov/investigations/process/Pages/default.aspx

NTSB On Site "Go Team"

- Investigator-in-Charge (IIC) – a senior investigator with years of NTSB and industry experience
  - NTSB investigators
  - Board member, who acts as official spokesperson;
  - Public affairs officer, who facilitates media briefings and responds to press inquiries
  - Transportation disaster assistant specialist

- NTSB investigators are specialists in a clearly defined area of the investigation such as:
  - OPERATIONS, STRUCTURES, SYSTEMS, ATC:
    - Accident flight history, crewmembers' duties prior to the crash; airframe analysis, vehicle systems, etc.
  - HUMAN PERFORMANCE:
    - Study crew performance, before-the-accident factors possibly involved in human error, e.g., fatigue, training, etc
  - SURVIVAL FACTORS:
    - Document impact forces and injuries, evacuation, community emergency planning and all crash-fire-rescue efforts.

"Go Team" Working Groups

Each NTSB investigator heads a working group in an area of expertise

- Groups are staffed by representatives of "parties" to the investigation
- Aviation accident-parties include FAA, airline, pilots' and flight attendants' unions, airframe and engine manufacturers
- Surface accident investigations, fewer working groups, same team technique
- Locomotive engineers, signal system specialists and track engineers head working groups at railroad accidents.
- Highway crashes include a truck or bus mechanical expert and a highway engineer.
- Weather, human performance and survival factors specialists respond to all accidents

- Working groups remain at the accident scene for a few days to several weeks.
- Work at Washington HQ forms the basis for analysis and drafting of report.

The "Party" System

- Board designates organizations as parties to its investigations
  - FAA, by law, is automatically designated a party
  - NTSB has complete discretion to designate parties
  - Only organizations that can provide expertise are granted party status
  - Only persons who can provide the Board with needed technical or specialized expertise are permitted to serve on the investigation
  - Persons in legal positions are not allowed
- All party members report to the NTSB
- Each investigative group chairman prepares a factual report for public docket

https://www.ntsb.gov/investigations/process/Pages/partysystem.aspx

Being Part of the NTSB Go Team

"It always happens when you least expect it."

Robert Sumwalt (right)

- Former NTSB Chair and Board Member;
- Associate Professor and Executive Director, Center for Aviation and Aerospace Safety,
  Embry Riddle Aeronautical University

Excerpts from "Being Part of the NTSB Go Team," Aviation Week, Robert Sumwalt, June 09, 2022

- Go Team schedule runs from 5pm on Monday to 5pm the following Monday.
- Phone may ring or receive a text to notify you of an upcoming call with the Response Operations Center (ROC) in 15 minutes.
- In 15 years as an NTSB board member, dispatched on 36 Go Team accidents.
- As Chairman, involved in numerous other calls and meetings at all hours of the day where we made decisions about our level of response."
- "Launch decision starts with "Chairman's Call" where key NTSB officials discuss known details of the situation and decide whether a full Go Team will be sent and whether board member will accompany the team.

Excerpts (cont.)

- We get to the accident site however we can. Often, FAA provides use of its aircraft. If not available, we fly on airlines.
- "FAA's plane could work around our schedule and take us to airports not served by air carriers. Accidents don't always happen close to airports served by airlines."
- Initial media briefing would provide footage that media could show throughout the day until we held a more detailed briefing upon arrival.
- Board member on scene would conduct media briefings, meet with families and elected officials.
- "Meeting with family members was hardest part of my job. Only solace I could provide was that NTSB was going to find out what happened, so others didn't have to go through what they were going through."

NTSB Safety Recommendations

- Issue recommendations to the organizations best able to take corrective action in the specific transportation area.
  - Examples : US DOT and its modal administrations, the Coast Guard, other federal and state agencies, manufacturers, operators, labor unions, and industry and trade organizations
- Monitor the progress of action to implement each recommendation until it is closed, which usually takes several years.
- Case Analysis and Reporting Online (CAROL)
  - Online search tool for investigations and recommendations across all modes.

NTSB's "Most Wanted List 1990-2023"

(see the MWL Archive)

Aviation

Require and Verify the Effectiveness of Safety Management Systems in all Revenue Passenger-Carrying Aviation Operations
Install Crash-Resistant Recorders and Establish Flight Data Monitoring Programs (video recorders)

Highway

Implement a Comprehensive Strategy to Eliminate Speeding-Related Crashes
Protect Vulnerable Road Users through a Safe System Approach
Require Collision-Avoidance and Connected-Vehicle Technologies on all Vehicles
Eliminate Distracted Driving

Marine

Improve Passenger and Fishing Vessel Safety

Rail, Pipeline, and Hazardous Materials

Improve Pipeline Leak Detection and Mitigation
Improve Rail Worker Safety

NTSB - Information Format

Uses docket system

https://data.ntsb.gov/Docket/Forms/recentlypublisheddockets

https://data.ntsb.gov/Docket/Forms/searchdocket

Similar to federal courts

- Federal courts create and maintain a case file which contains a docket sheet and all documents filed in a case.
- Case files and court records can be found on (public access to court electronic records) PACER.gov. https://www.uscourts.gov/court-records

Investigation example - Conception

75-foot dive boat; 33 passengers, 6 crew

- September 2, 2019, Conception dive boat caught fire at 3 am, burned to the waterline, and sank less than 100 feet from shore.
  - Anchored off Santa Cruz island
- 34 people asleep below deck in the bunkroom were trapped.
  - None survived.

NTSB investigation page

Example

NTSB Opens Public Docket on Conception Dive Boat Fire

- https://www.ntsb.gov/news/press-releases/Pages/NR20200916.aspx

Initial Report (20 Oct 2020)

- https://www.ntsb.gov/news/events/Documents/2020-DCA19MM047-BMG-abstract.pdf

Final Report

- https://www.ntsb.gov/investigations/AccidentReports/Reports/MAR2003.pdf

After the Investigation, What Happens?

- Probable cause was failure of the operator, Truth Aquatics, to provide effective oversight of its vessel and crewmember operations allowing fire to grow undetected.
- Example of need for rigorous fire safety standards for small passenger vessels
- Lack of USCG regulatory requirement for smoke detection in all accommodation spaces and inadequate emergency escape arrangements from the vessel's bunkroom contributed to the undetected growth of the fire
- 21 open NTSB recommendations to USCG focused on improving passenger vessel safety.
- Reiterated recommendation to USCG to require safety management systems (SMS) on U.S. flag passenger vessels.
- Congress codified these safety recommendations into law, mandated USCG implement recommendations in the Elijah E. Cummings Coast Guard Authorization Act of 2020 as part of the National Defense Authorization Act.

https://www.ntsb.gov/investigations/AccidentReports/Reports/MAR2003.pdf

https://www.tsb.gc.ca/eng/enquetes-investigations/index.html

Other Transportation Investigative Entities

- Federal Railroad Administration
  https://railroads.dot.gov/railroad-safety/accident-data-reporting-and-investigations

- Federal Motor Carrier Safety Administration
  https://ai.fmcsa.dot.gov/NewEntrant/MC/Content.aspx?nav=Accidents

- US Coast Guard
  https://cgmix.uscg.mil/IIR/Default.aspx

NASA Mishap Investigation Team (MIT)

A NASA mishap is an unplanned event resulting in at least one of the following:

- Occupational injury or occupational illness to NASA or non-NASA personnel caused by NASA operations
- Destruction of, or damage to, NASA property, public or private property, including foreign property, caused by NASA operations or NASA-funded research and development projects
- NASA mission failure before the scheduled completion of the planned mission
- "Close call"
  - no injury or only minor injury requiring first aid, no damage or minor damage (less than $20,000) to equipment or property or both, but possesses potential to cause a mishap

A Quick Reference Guide for Mishap Investigations (2020)

References

- Kenneth, L. Carper, Forensic Engineering, Second Edition, 2001. Chapter 7